Python-Based Pre-Harvest Pomegranate Fruit Detection, Yield Estimation and Production Assessment

$0.3cm]An M.Tech Technical Literature Review and Proposed Architectur$

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October 2026

Abstract

Pre-harvest agricultural yield and total production estimation plays a vital role in modern precision horticulture, supply-chain planning, and farm financial management. Focusing on pomegranate (*Punica granatum L.*) orchards, this literature review critically synthesizes seven foundational research studies spanning on-tree computer vision phenotyping, temporal tracking, commodity price forecasting, contract farming game theory, and real-options lease valuation. While state-of-the-art deep learning architectures achieve up to 98.0% growth stage classification accuracy (CRnet), 0.922 mAP in multi-stage fruit detection at 29.8 FPS on mobile devices (YOLO-Granada), and ~ 1.0 cm MAE in 3D physical diameter estimation using RGB-D depth projection, existing solutions remain restricted to 2D/3D image-level metrics. A critical gap persists between visual fruit sensing and total biomass estimation: existing systems do not convert visible fruit counts and projected dimensions into calibrated tree biomass (kg/tree) and total orchard production while accounting for canopy foliage occlusion and error propagation. To address this limitation, this report proposes a Python-based system architecture integrating multi-view video de-duplication, 3D volumetric mass allometrics, and a canopy visibility correction factor to deliver field-grounded pre-harvest yield forecasts accompanied by 95% confidence error bounds.

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1 Introduction: The Gap Between Sensing and Valuation

Modern agricultural management relies increasingly on precision decision-support tools to optimize harvest timing, labor deployment, market distribution, and financial contracting. Pomegranate (*Punica granatum L.*) represents a high-value horticultural crop of major economic importance in Maharashtra, India, and semi-arid regions globally. However, orchard management remains heavily reliant on subjective manual inspection, sporadic sampling, and visual heuristics. This traditional paradigm suffers from high labor expenses, susceptibility to human sampling bias, and an inability to provide timely, accurate forecasts of total standing crop biomass prior to harvest.

In recent years, computer vision (CV) and deep learning (DL) have demonstrated remarkable capabilities in agricultural phenotyping. Convolutional Neural Networks (CNNs) and lightweight object detectors are capable of identifying individual fruits under complex field lighting, classifying developmental growth stages, and measuring physical dimensions using non-contact 3D depth sensors. Concurrently, agricultural economists have developed sophisticated mathematical models for commodity price forecasting, contract farming risk sharing, and real-options fruit tree lease pricing.

Despite these parallel advancements, a fundamental technical disconnect exists between field sensing and economic valuation. Computer vision algorithms typically terminate their execution pipeline at frame-level object detection, 2D bounding box counts, or 3D physical diameter measurements. They do not convert optical observations into unique non-redundant fruit counts per tree, individual fruit mass in grams, or total orchard production in kilograms or metric tons. Conversely, economic valuation and contract optimization models operate on abstract mathematical probability distributions (such as Poisson or Gaussian yield assumptions) completely detached from physical on-tree field observations.

This literature review presents a rigorous technical synthesis of seven authoritative research papers. It categorizes existing solutions into on-tree phenotyping, temporal tracking, and economic decision modeling, explicitly identifying the technical friction points that prevent current sensing technologies from estimating total crop production. Finally, it outlines a proposed Python-based system architecture designed to bridge the count-to-mass calibration gap through multi-view de-duplication, 3D volumetric mass allometric modeling, foliage occlusion correction, and statistical error propagation.

2 On-Tree Phenotyping: Isolated Solutions and Limitations

2.1 Naseer et al. (2024) — Pomegranate Growth Stage Detection

Full Citation: A. Naseer, M. Amjad, A. Raza, K. Munir, N. A. Samee, and M. A. Alohali, A Novel Transfer Learning Approach for Detection of Pomegranates Growth Stages," *IEEE Access*, vol. 12, pp. 127880–127894, 2024. [DOI Conflict: Publication citation line states 10.1109/ACCESS.2024.3365356, while internal front-matter states 10.1109/ACCESS.2023.1120000].

- **Dataset & Hardware:**
  - Mendeley Data public pomegranate growth-stage image collection comprising 5,857 cropped images.
  - Class distribution: Bud (1,009), Flower (1,597), Early-Fruit (772), Mid-growth (1,408), and Ripe (1,071).
  - Data split: 80% training and 20% testing ratio; Synthetic Minority Over-sampling Technique (SMOTE) applied to balance class frequencies.

  - Hardware: Executed on Google Colab GPU environment with Intel Xeon CPU, 13 GB RAM, and 90 GB storage.
  - Geographic site, orchard identities, capture dates, lighting conditions, and independent farm sample size: *Not reported in the paper*.

- **Core Architecture/Method:**
  - Proposed **CRnet** architecture combining 2D Convolutional Neural Networks for spatial feature extraction with Random Forest (RF) probability estimation.
  - Input images resized to $244 \times 244 \times 3$ RGB pixels; 2D-CNN extracts a 600-dimensional spatial feature vector (5857×600 matrix).
  - Spatial feature vector passed to Random Forest to generate class probability features ($y_i|X$), creating a novel probabilistic feature space.
  - Evaluated secondary classifiers on transformed features: Random Forest (RF), K-Neighbors Classifier (KNC), Logistic Regression (LR), Gaussian Naive Bayes (GNB), and standard 2D-CNN baseline.
  - Implemented in Python 3 using TensorFlow, Keras, and scikit-learn libraries with 10-fold cross-validation.

- **Key Metrics:**
  - CRnet + Random Forest achieved 98.03% overall classification accuracy (std dev $\pm 0.43\%$), 0.98 Precision, 0.98 Recall, and 0.98 F1-score across all 5 growth stages.
  - Class-wise 10-fold accuracy: 98.0% across Bud, Flower, Early-Fruit, Mid-growth, and Ripe categories (Table 6 in paper).
  - Other classifiers on CRnet features: Logistic Regression = 97.96%, Gaussian Naive Bayes = 97.70%, K-Neighbors = 97.63%.
  - Standard 2D-CNN baseline achieved only 55.0% accuracy with severe misclassification between Bud and Flower stages.
  - Model construction runtime: Random Forest = 3.741 s, KNC = 0.039 s, LR = 0.202 s, GNB = 0.005 s, CNN = 2.015 s.

- **Additional Technical Details:**

– Hyperparameter settings: RF `max_depth` = 300, `n_estimators` = 300, `min_samples_leaf` = 1; KNC `n_neighbors` = 5; LR L2 penalty =1.0.$SMOTEoversamplingeffectivelysolvedtrainingbiascausedbycl$

Critical Analysis and Limitations The CRnet framework developed by Naseer et al. demonstrates exceptional classification performance on pre-cropped single-organ images, effectively distinguishing subtle visual transitions between early buds, flowers, and developing pomegranate fruits. Transforming CNN spatial features into Random Forest probabilistic representations overcomes the traditional feature degradation observed in shallow CNN baselines. However, from a yield estimation perspective, the study operates under severe methodological constraints. The system assumes pre-cropped, centered organ images and possesses zero object detection or spatial localization capabilities; it cannot identify or segment pomegranates within full, uncropped orchard canopy scenes. Furthermore, the paper provides no fruit

counting, physical dimension measurement, or mass conversion mechanisms. Because the geographic location, farm identities, and camera acquisition setups are not reported, the model's robustness against variable outdoor field lighting and foliage shading remains unverified. CRnet functions effectively as a growth-stage verifier, but cannot independently perform crop load or yield estimation.

Figure 2: Schematic Topology of CRnet Architecture (Naseer et al., 2024)

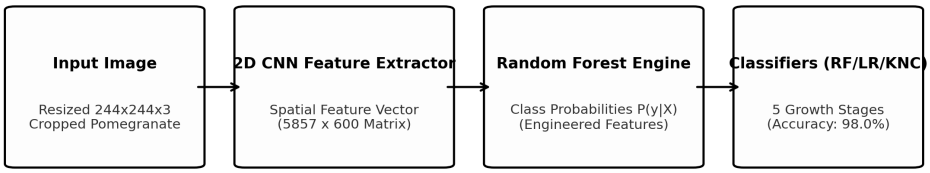


Figure 1: Schematic Topology of CRnet Architecture combining 2D-CNN Spatial Feature Extraction with Random Forest Probabilistic Feature Engineering (Adapted from Naseer et al., 2024).

2.2 Zhao et al. (2024) — YOLO-Granada Lightweight Fruit Detection

Full Citation: J. Zhao, C. Du, Y. Li, M. Mudhsh, D. Guo, Y. Fan, X. Wu, X. Wang, and R. Almodfer, YOLO-Granada: A Lightweight Attentioned YOLO for Pomegranates Fruit Detection," *Scientific Reports*, vol. 14, art. 16848, 2024. DOI: 10.1038/s41598-024-67526-4.

- **Dataset & Hardware:**
  - Mendeley Data pomegranate growth-period dataset consisting of 5,857 full orchard images.
  - Data split: 4,685 training (80%), 585 validation (10%), and 587 test images (10%), containing 1,105 annotated target instances in the test set.
  - Evaluation hardware: Desktop CPU platform powered by Intel Core i5-10400F CPU (no GPU acceleration used during CPU benchmark speed tests).
  - Mobile deployment platform: Commercial Android smartphone running NCNN C++ inference framework.
  - Specific orchard location, capture dates, tree count, and cultivar details: *Not reported in the paper*.
- **Core Architecture/Method:**
  - **YOLO-Granada** architecture based on YOLOv5s baseline object detector.
  - Backbone network replaced with lightweight ShuffleNetv2 utilizing grouped convolutions and channel shuffling to reduce computational parameters.
  - Convolutional Block Attention Module (CBAM) integrated into the feature fusion neck to enhance spatial and channel-wise feature representation for small targets.
  - Multi-scale feature outputs at 80×80, 40×40, and 20×20 feature map resolutions.

  - Model exported via ONNX and simplified using NCNN framework for real-time mobile C++ execution on Android.
  - Trained using PyTorch 1.11, CUDA 11.2, Adam optimizer, L2 regularization, and confidence threshold of 0.001 with NMS IoU threshold of 0.6.

- **Key Metrics:**
  - **Model Lightweighting (vs YOLOv5s baseline):** Parameters = 3.84 M (54.7% of YOLOv5s), GFLOPs = 8.1 (51.3% of YOLOv5s), Model File Size = 7.74 MB (56.3% of YOLOv5s).
  - **Detection Accuracy:** mAP@0.5 = 0.922 (only a minor 0.7% drop from baseline YOLOv5s mAP of 0.929), Precision = 0.929, Recall = 0.833, F1-score = 87.84%.
  - **Class-wise mAP@0.5:** Bud = 0.872, Flower = 0.945, Early-fruit = 0.910, Mid-growth = 0.899, Ripe = 0.987.
  - **Inference Speed:** 8.66 FPS on Intel Core i5 CPU (17.3% faster than YOLOv5s at 7.19 FPS); 29.7–29.8 FPS real-time execution on Android video streams.

- **Additional Technical Details:**
  - Android deployment built using Gradle 7.2.0, JDK 18, targeting Android 10 with multithreaded stream processing.
  - Ablation studies proved that CBAM attention recovered detection accuracy lost during ShuffleNetv2 backbone compression.

Critical Analysis and Limitations Zhao et al. successfully address the computational deployment bottleneck of deep learning in precision agriculture by engineering a lightweight, attention-guided detector capable of real-time execution on low-cost mobile hardware. The integration of ShuffleNetv2 and CBAM achieves a remarkable 43.7% reduction in model size (7.74 MB) while maintaining a high detection accuracy of 0.922 mAP across five growth stages. Nevertheless, YOLO-Granada remains strictly a single-frame 2D object detector. The model outputs 2D bounding boxes, class labels, and confidence scores within isolated image frames. It lacks multi-frame temporal tracking or video de-duplication logic; consequently, traversing an orchard row with a camera causes repeated double-counting of the same fruit across consecutive frames. Furthermore, 2D bounding boxes provide no 3D spatial depth or physical scale, making it impossible to measure physical fruit diameter, volume, or weight. The authors also explicitly acknowledge detection failures under extreme sunlight contrast and severe foliage occlusion.

2.3 Devanna et al. (2025) — Automated Robot Sensing & 3D Fruit Sizing

Full Citation: R. P. Devanna, F. Vicino, S. P. Garofalo, G. A. Vivaldi, S. Pascuzzi, G. Reina, and A. Milella, Automated On-Tree Detection and Size Estimation of Pomegranates by a Farmer Robot," *Robotics*, vol. 14, art. 131, Sep. 2025. DOI: 10.3390/robotics14100131.

- **Dataset & Hardware:**
  - Field dataset collected in a commercial pomegranate orchard in Grottaglie, Apulia, southern Italy, in October 2021 (just prior to harvest).

Figure 3: YOLO-Granada Network Topology with ShuffleNetv2 & CBAM (Zhao et al., 2024)

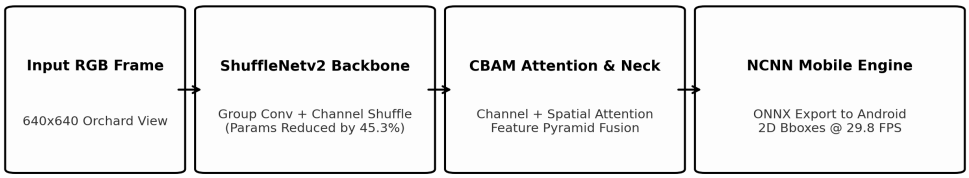


Figure 2: YOLO-Granada Network Architecture featuring ShuffleNetv2 Backbone, CBAM Attention Modules, and NCNN Mobile Export Engine (Adapted from Zhao et al., 2024).

  - Mobile sensing platform: Tracked ground robot (Polibot" / Farmer Robot) equipped with RTK GNSS/INS and a consumer-grade Intel RealSense D435 RGB-D camera mounted 1.0 m above ground.
  - Camera setup: Lateral row perspective, camera-to-tree distance $\sim$ 1.5 m, recording at 6 FPS (280 $\times$720 resolution).
  - Data scale: 945 RGB-D frames captured across 15 pomegranate trees along an orchard row (3.0 m tree spacing).
  - Ground-truth sizing dataset: 96 pomegranates randomly harvested from the 15 trees for manual digital caliper measurements.

- **Core Architecture/Method:**
  - **Segmentation:** Semi-supervised 3-stage transfer learning fine-tuning DeepLabv3+ with a ResNet18 backbone. Stage 1 trained on controlled monochrome background images; Stage 2 refined under variable indoor lighting; Stage 3 fine-tuned on field orchard images.
  - **2D Geometry Modeling:** Connected component blob extraction followed by 2D circumscribed ellipse fitting. Applied an eccentricity threshold filter to discard non-round, distorted, or severely occluded fruit contours.
  - **3D Point Cloud Projection:** Projected the 2D ellipse major and minor axis vertices (poles) onto the spatially aligned RealSense D435 3D depth point cloud.
  - **Depth Filtering:** Applied a standard deviation depth outlier filter (=1σ distance from mean blob depth) to eliminate noisy depth pixels at fruit silhouettes.
  - **Physical Size Computation:** Calculated 3D Euclidean distance between projected 3D pole coordinates to yield physical equatorial diameter $(, cm) and polar diameter(,$ cm).

- **Key Metrics:**
  - **Segmentation Performance:** Precision = 93.33%, Recall = 81.49%, F1-score = 86.42%, processing speed = 0.15 s/frame ($\sim$ 6.6 FPS).
  - **Physical Sizing Accuracy (vs 96 Caliper Measurements):**

* Equatorial Diameter () : $Calipermean = .91 \pm 0.85$ cm; Robot estimated mean = .15 ± 1.31 cm. MAE = 1.10 cm, RMSE = 1.35 cm, Standard Deviation = 0.78 cm, MAPE = 12.4%.
* Polar Diameter () : $Calipermean = .89 \pm 0.88$ cm; Robot estimated mean = .62 ± 1.37 cm. MAE = 1.05 cm, RMSE = 1.31 cm, Standard Deviation = 0.79 cm, MAPE = 13.3%.

- **Additional Technical Details:**
  - Operates completely target-free without requiring physical calibration markers in the orchard scene.
  - RGB field-of-view $9^\circ \times 42^\circ$; depth field-of-view $7^\circ \times 58^\circ$; effective depth range 0.3–3.0 m.

Critical Analysis and Limitations Devanna et al. provide a major technical breakthrough by demonstrating calibration-free, non-contact 3D physical fruit sizing directly on trees in natural field settings. By combining semi-supervised 2D semantic segmentation with RGB-D point cloud projection, the Polibot system achieves physical diameter estimations with an MAE of approximately 1.0 cm. The 3-stage transfer learning pipeline drastically reduces manual annotation labor. However, from a total yield estimation perspective, the methodology contains critical gaps. Most importantly, the system explicitly filters out and discards occluded, overlapping, or non-round fruits using an eccentricity threshold. While this filtering is necessary to ensure clean ellipse fitting for 3D sizing, it completely eliminates partially hidden fruits from the measurement pipeline, introducing a severe negative bias in total fruit count. Furthermore, the authors measure 3D physical diameters (, PD$incm)butdonotformulateanallometricequationtoconvertgeometri$
$Dframeswithoutmulti-viewvideotracking, leavingtotaltreefruitloadandorchardproductionuncalculated.$

Figure 5: Polibot Mobile Sensing & 3D Ellipse Depth Projection Pipeline (Devanna et al., 2025)

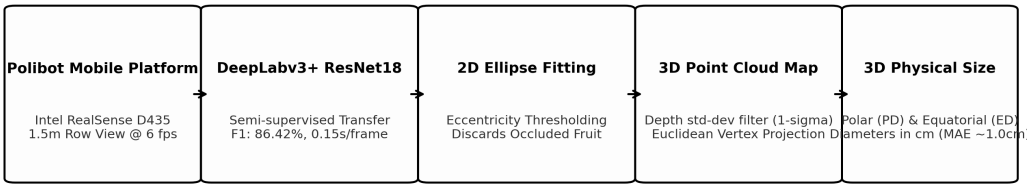


Figure 3: Polibot Mobile Robot Platform, DeepLabv3+ Segmentation, Ellipse Fitting, and 3D Depth Point Cloud Projection Pipeline for Physical Diameter Sizing (Adapted from Devanna et al., 2025).

3 Temporal Tracking and the Time Dimension

3.1 Giménez-Gallego et al. (2025) — On-Tree Tracking & Harvest Date Prediction

Full Citation: J. Giménez-Gallego, J. Martínez-del-Rincon, P. J. Blaya-Ros, H. Navarro-Hellín, P. J. Navarro, and R. Torres-Sánchez, Fruit Monitoring and Harvest Date Prediction Using On-Tree Auto-

matic Image Tracking," *IEEE Transactions on AgriFood Electronics*, vol. 3, pp. 56–68, 2025. [DOI: *Not reported in the paper*].

- **Dataset & Hardware:**
  - Field study conducted over two growing seasons (2021 and 2022, June to October) in a commercial 'Wonderful' pomegranate orchard in Mazarrón, Murcia, Spain.
  - Sensor hardware: 9 autonomous IoT camera nodes equipped with Raspberry Pi Zero W, 5-Megapixel RGB camera modules, solar panels, and LTE Wi-Fi routers.
  - Camera mounting: Overhead zenithal (top-down) perspective mounted on agricultural posts above the tree canopy.
  - Image capture frequency: 5 images daily (9 A.M., 11 A.M., 1 P.M., 3 P.M., 5 P.M. GMT+2); 9 A.M. daily capture selected for tracking.
  - Data scale: Segmentation dataset of 400 images (624 $\times$3472 resized to 40 $\times$480); Harvest regression dataset of 68 monitored fruit time series comprising 4,850 daily measurement observations ($\sim$ 71 monitored days per fruit).
- **Core Architecture/Method:**
  - **Instance Segmentation:** Mask R-CNN with ResNet50-FPN backbone pretrained on COCO, fine-tuned on 400 annotated pomegranate images (70% train / 15% val / 15% test).
  - **Multi-Target Tracking:** Linear programming multi-target tracking algorithm linking segmented fruit instance IDs across consecutive Days of Year (DOY).
  - **Feature Extraction:** Extracted median offset Hue $()inHSVcolorspace, normalized2-, 4-, and8-binoffsetHuehistograms(hist2, H_{hist4}, H_{hist8})$, fitted circle radius in pixels $(), andelapseddaysfromfull-bloomreferencedate()$.

– **Harvest Date Regression:** XGBoost Regressor predicting remaining days until commercial harvest $().$**Time Window Subseries (TWS):**$Evaluatedslidingtimewindows(1, 3, 5, 10, 20, 30, 40days)withrandomizedt$

- **Key Metrics:**
  - **Mask R-CNN Segmentation:** F1-score = 0.817, Intersection over Union (IoU) = 0.784.
  - **Image vs In-Field Manual Ground Truth Correlation:** Circle-fitted pixel size vs manual caliper $^2 = 0.80$ (median $^2 = 0.85$); HSV offset Hue vs portable colorimeter $^2 = 0.70$.
  - **Harvest Prediction (Best Model: Feature Vector 9 [Color + Size + ED], TWS = 1 day):**
    - Median Error = 3.5 days, Mean Error = 4.5–5.2 days, RMSE = 6.2 days, Coefficient of Determination $^2 = 0.88$.
    - Prediction Accuracy: Week Accuracy ($\pm$3.5 days) = 42%, Fortnight Accuracy ($\pm$7.5 days) = 74%, Month Accuracy ($\pm$15 days) = 98%.
- **Additional Technical Details:**
  - Low-cost IoT node build expense < 50 per camera unit.

– Including full-bloom reference date $()eliminatedtherequirementforlongtemporalimagesequences, allowingtop$

Critical Analysis and Limitations Giménez-Gallego et al. demonstrate an innovative, ultra-low-cost IoT system for continuous temporal tracking of fruit growth and color maturation under field conditions. By integrating Mask R-CNN segmentation with a linear programming tracking algorithm and XGBoost regression, the system successfully forecasts optimal commercial harvest dates with a median error of 3.5 days. The discovery that incorporating full-bloom timing $()enablesaccuratesingle-dayharvestpredictionisofsignificantpracticalvalue.However,thestudypossessesdistinctmethodologicallimit$ $downcameraperspectiveobservesonlytheupperexteriorcanopy,capturingapproximately4top-exposedfruitspe$ which cannot be converted into physical metric dimensions (cm) or physical mass (grams) because camera-to-fruit distance is uncalibrated. Finally, ground-truth harvest labels reflect commercial grower picking logs, which contain operational noise due to labor scheduling and market conditions. The paper predicts harvest timing for individual visible fruits, but does not calculate total tree crop load or orchard production volume.

Figure 4: Zenithal IoT Camera Monitoring Setup & XGBoost Tracking Pipeline (Giménez-Gallego et al., 2025)

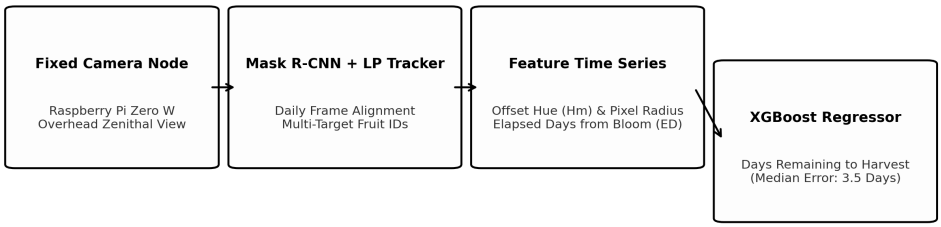


Figure 4: Zenithal IoT Camera Setup, Mask R-CNN + Linear Programming Multi-Target Tracking, and XGBoost Harvest Date Prediction Pipeline (Adapted from Giménez-Gallego et al., 2025).

4 Economic Context and Decision Uncertainty

4.1 Mahajan et al. (2026) — Agricultural Commodity Price Forecasting

Full Citation: P. Mahajan, A. A. Muley, and M. R. Fegade, Forecasting Agricultural Commodity Prices Using Machine Learning Models," *Discover Informatics*, vol. 1, art. 15, pp. 1–16, 2026. DOI: 10.1007/s44564-026-00015-0.

- **Dataset & Hardware:**
  - AGMARKNET daily mandi-level market price records from 2001 to October 2025 across wholesale markets in Maharashtra, India.
  - Dataset scale: 1.6 million daily market records aggregated into monthly modal price time series across 9 major commodities (Arhar, Bajra, Cotton, Maize, Paddy, Sesamum, Soyabean, Sunflower, Wheat). Individual record counts range from 27,832 to 395,294 per crop.
  - Evaluation window: Sequential expanding-window testing from January 2021 to October 2025 (58 testing months).
- **Core Architecture/Method:**

  - Machine learning time-series regression pipeline predicting one-month-ahead modal price ($\hat{P}_t$, INR/quintal).
  - Logarithmic target transformation = $\log(1 + P_t)$ and outlier capping using 0.1st and 99.9th percentiles fitted strictly on prior training history to eliminate data leakage.
  - Engineered features: Lagged prices (-1, t-2, t-3, t-6, t-9, t-12, t-18, t-24), $rollingwindowmeans(3, 6, 12, 24mon$
  - Evaluated tree-based ensemble regressors: Random Forest (RF), Extra Trees (ET), Histogram Gradient Boosting (HGB), and XGBoost (XGB).
  - Benchmark models: 1-month Naive persistence baseline ($\hat{P}_t = P_{t-1}$) and 12-month Seasonal Naive baseline ($\hat{P}_t = P_{t-12}$).

- **Key Metrics:**
  - **1-Month Naive Benchmark Superiority:** The simple 1-month Naive persistence benchmark achieved the lowest Mean Absolute Percentage Error (MAPE) across **ALL 9 commodities** (Paddy: 2.30%, Wheat: 2.46%, Soyabean: 3.05%, Bajra: 3.40%, Cotton: 3.84%, Sesamum: 4.60%, Maize: 4.62%, Arhar: 4.74%, Sunflower: 4.91%).
  - **Machine Learning Performance:** ML models consistently outperformed the 12-month Seasonal Naive benchmark, but **failed to beat the 1-month Naive persistence benchmark** for any commodity (Best ML MAPEs: Paddy RF = 4.58%, Wheat RF = 5.45%, Soyabean XGB = 8.60%, Bajra RF = 5.59%, Cotton RF = 10.41%, Sesamum ET = 11.74%, Maize RF = 7.07%, Arhar XGB = 14.18%, Sunflower XGB = 10.71%).
  - **Expanding Retraining Impact:** Sequential expanding-window retraining drastically improved ML accuracy over fixed-window models (e.g., Arhar ML MAPE dropped from 28.85% to 14.18%).

- **Additional Technical Details:**
  - Implemented in Python using scikit-learn and XGBoost libraries.
  - Demonstrates that historical price time series exhibit strong short-term inertia in agricultural spot markets.

Critical Analysis and Limitations Mahajan et al. present a rigorous, leakage-safe empirical evaluation of price forecasting models across agricultural spot markets in Maharashtra. Their finding that sophisticated machine learning models relying solely on past prices cannot surpass a simple 1-month Naive persistence benchmark provides a vital baseline insight. However, the study suffers from a major structural limitation: it relies exclusively on historical price lag features while completely excluding exogenous physical drivers such as weather events, market arrivals, government MSP policy, and—most importantly—crop production and yield data. Price series alone cannot anticipate sudden market supply shocks caused by seasonal yield fluctuations. Furthermore, the paper analyzes 9 field and oilseed crops across wholesale mandis, but does not model pomegranate prices or farm-gate transactions. This study demonstrates that price forecasting cannot break through persistence baselines without incorporating advance pre-harvest crop yield and production estimates.

4.2 Nematollahi et al. (2025) — Contract Farming Under Uncertainty

Full Citation: M. Nematollahi, A. Guitouni, J. Heydari, and E. M. Gerbrandt, Win–win Contract Farming in Dual-Channel Agribusiness Supply Chains Under Yield, Quality, and Price Uncertainty," *International Journal of Production Economics*, vol. 286, art. 109635, pp. 1–16, 2025. DOI: 10.1016/j.ijpe.2025.109635.

Figure 6: Mandi Modal Price Forecasting & Expanding-Window Protocol (Mahajan et al., 2026)

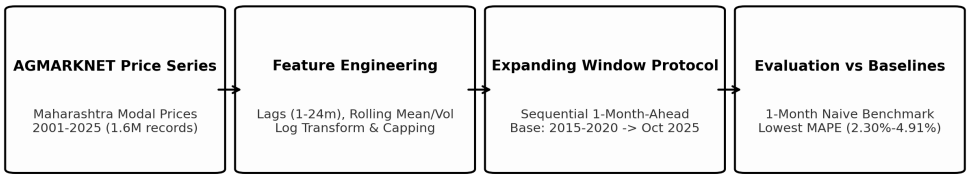


Figure 5: AGMARKNET Mandi Price Time Series, Feature Engineering, and Sequential Expanding-Window Evaluation Protocol (Adapted from Mahajan et al., 2026).

- **Dataset & Hardware:**
  - Numerical parameterization based on a representative 40-acre commercial blueberry farm in Fraser Valley, British Columbia, Canada.
  - Historical price data: USDA weekly fresh blueberry market price reports (2000–2020) and British Columbia agricultural production statistics.
  - Source data scale: Analytical mathematical model; field image dataset and computer vision acquisition: *Not reported in the paper*.
- **Core Architecture/Method:**
  - Game-theoretic Stackelberg leader-follower optimization modeling contract coordination between agricultural growers and distributors in dual fresh/processed channels.
  - Evaluated 3 contract structures: (1) No-contract spot market selling, (2) Traditional penalty-based contract farming, and (3) Incentive-based contract farming with production cost-sharing (α_{sc}).

– Stochastic variables modeled: Harvest yield rate ($\xi \sim \mathcal{N}(\mu_g = 15000, \sigma_g = 4000)$ lb/acre), fruit quality diameter ($\theta_f \sim \mathcal{N}(\mu_h = 7.92, \sigma_h = 0.69)$ mm, with minimum fresh threshold $\Theta_f = 8.0$ mm yielding 51% fresh qualification), and fresh spot market price ($\tilde{p}_f^s \sim \mathcal{N}(2.32, 0.62)\ /lb).Solvedviabackwardinductio$
$formanalyticalexistenceboundariesforwin - winsupplychaincoordination.$

- **Key Metrics:**
  - **Supply Chain Profitability (40-acre Representative Farm):**
    * No-Contract Baseline: Total SC Profit = $392,789 (FarmerProfit$ =392,789, Distributor = $0).Penalty-BasedContract : TotalSCProfit$ =474,018 (Farmer = $374,634, Distributor$ =99,384, Wholesale Price = $1.328/lb$).

* Incentive-Based Contract ($\alpha_{sc} = 0.715$ cost share): Total SC Profit = $554,673 (Farmer$ =424,039, Distributor = $130,634, WholesalePrice$ =1.062/lb).
  - **Profit Improvement:** Incentive contracting increased total supply chain profit by **17.0% over penalty contracting** and **41.2% over the no-contract spot baseline**.

- **Additional Technical Details:**

– Proves that production cost-sharing ($\alpha_{sc} = 71.5\%$) eliminates grower default risk and optimizes fresh market volume commitment (= 305,228 lbs).

Critical Analysis and Limitations Nematollahi et al. provide a compelling game-theoretic proof demonstrating that collaborative, incentive-based contract farming with production cost-sharing significantly outperforms punitive contracts under yield, quality, and price risks. Their analytical framework illustrates how pre-harvest quality thresholds (Θ_f, fruit diameter) and volume commitments directly dictate supply-chain surplus. However, the study operates entirely as a mathematical optimization model. Crop yield (ξ) and fruit quality proportion (θ_f) are modeled as assumed, aggregate Gaussian probability distributions completely detached from real-time field computer vision or spatial crop sensing. The model contains no image processing, fruit detection, or in-field diameter measurement capabilities. Furthermore, the model is parameterized specifically for Fraser Valley blueberries and assumes complete information symmetry between growers and distributors. This work highlights the economic value of yield and quality estimation, but relies entirely on theoretical yield assumptions.

4.3 Abdurakhman et al. (2025) — Valuing Fruit Tree Lease Contracts

Full Citation: A. Abdurakhman, A. Sihabuddin, K. C. Wijaya, E. Purnaningrum, and D. A. I. Maruddani, Valuing Fruit Tree Lease Contracts Under Uncertainty: A Probabilistic Framework for Fair Pricing," *Iraqi Journal for Computer Science and Mathematics*, vol. 6, no. 3, art. 39, pp. 1–12, 2025. DOI: 10.52866/2788-7421.1308.

- **Dataset & Hardware:**

  – Preliminary field survey observations and farmer interviews conducted in December 2024 across rural durian tree leasing communities in Yogyakarta and East Java, Indonesia.

  – Primary data scale: Parameterized survey data; exact interview sample size and raw image datasets: *Not reported in the paper.*

– Baseline model parameters: Mean yield $\lambda = 120$ fruits/tree, fruit price bounds IDR 30,000–50,000/fruit, maintenance cost = IDR 1,000,000$/tree/harvest, yieldthreshold = 100$ fruits, risk-free rate = 5%.

- **Core Architecture/Method:**

  – Real Options mathematical valuation framework for pricing long-term fruit tree lease contracts under dual yield and price uncertainty.

  – Developed and contrasted two real-option models: **Fixed-Time Discount (FTD)** model and **Dynamic-Time Discount (DTD)** model.

  – Stochastic variable modeling: Harvest yield $()modeledviaPoissondistribution(\sim$ Poisson$(\lambda))$; fruit market price $()modeledviaUniformdistribution(\sim$ Uniform$(a, b))$.

  – DTD model represents contract tenure $()asarandomvariablefollowingaGeometricdistribution(\sim$ Geom$(p))$, where contract duration dynamically extends if harvest yield falls below threshold $.Computedfairleaseprice()$ and expected net tenant profit ($\tilde{f}_T$) using Monte Carlo numerical simulations in Python.

- **Key Metrics:**

  - **Model Lease Pricing Comparisons (Baseline $\lambda = 120, K_N = 100$):** FTD Calculated Price = IDR 1,851.70/tree; DTD Calculated Price = IDR 1,877.55/tree (compared to empirical market baseline IDR 1,000,000/tree).
  - **Sensitivity Under Low Yield ($\lambda < K_N$, e.g., $\lambda = 80$ fruits):** DTD model dynamically adjusts lease price upward to IDR 7,430.92 to compensate for contract extension risk, whereas static FTD collapses to IDR 594.59.
- **Tenant Loss Risk:** Under static FTD, if yield drops 10 units below threshold , $tenantincursa6.5\%netfinancialloss$;

- **Additional Technical Details:**
  - Demonstrates that dynamic time discounting prevents financial disputes and contract defaults during severe low-yield years.

Critical Analysis and Limitations Abdurakhman et al. address an important problem in rural agricultural finance by formulating a real-options pricing framework for fruit tree lease contracts. By introducing dynamic time discounting linked to Poisson yield thresholds, their DTD model protects tenant farmers from severe financial losses during bad harvest years. However, the framework operates purely as a theoretical financial model. The key biological variable—mean harvest yield per tree (λ)—is modeled as an assumed Poisson random variable rather than being measured from standing crops in the orchard. The model contains no computer vision, remote sensing, or image processing capabilities. Furthermore, the authors acknowledge that assuming strict independence between fruit price and harvest yield is unrealistic in real agricultural markets. Parameterized for Indonesian durian trees, this framework provides valuable economic context for tree leasing, but offers no field-validated vision tools to measure actual standing yield.

5 Research Gaps: The Count-to-Mass Calibration Problem

5.1 The Distinction Between Detection, Counting, Diameter, and Mass

A rigorous synthesis of the reviewed literature reveals a fundamental technical disconnect between what existing computer vision algorithms output and what agricultural yield estimation requires. As summarized in Table 1, current phenotyping algorithms execute isolated tasks along the sensing pipeline, but fail to complete the transition to biomass estimation:

1. **Frame Detection vs. Unique Fruit Counting:** Object detectors such as YOLO-Granada (Zhao et al., 2024) output 2D bounding boxes and frame-level counts (detect).$However, single-framedetectionlacksspatialde$ $duplicationacrossmovingcameraframes.Traversingatreerowresultsinrepeateddouble-countingofidentic$ $levelcropload.$**Geometric Diameter vs. Physical Mass:**$Sizingalgorithmssuchas Polibot(Devannaetal., 2025)e$ in cm). However, geometric diameter is a linear spatial metric, whereas crop yield is defined by physical biomass or weight (kg/tree or tons/ha). Because fruit volume scales non-linearly with diameter, diameter measurements alone cannot quantify total yield without a calibrated volumetric mass allometric model (= f(ED$_i$, PD_i)).

5.2 The Canopy Occlusion and Visibility Barrier

Foliage occlusion represents the single greatest physical barrier to visual yield estimation in pomegranate orchards. Pomegranate trees possess dense canopy foliage, short branches, and interior fruit clusters

that hide a substantial fraction of fruits from exterior visual observation. The reviewed literature handles occlusion in two flawed ways:

- **Explicit Discard:** Sizing pipelines like Devanna et al. (2025) explicitly filter out and discard occluded, overlapping, or non-round fruits using an eccentricity threshold to ensure clean ellipse fitting. While necessary for 3D sizing, discarding occluded fruits removes them from the counting pipeline, causing severe underestimation of total tree fruit load.
- **Perspective Bias:** Fixed overhead monitoring systems like Giménez-Gallego et al. (2025) observe trees from a zenithal top-down view, capturing only top-exposed fruits (~ 4 fruits/tree) while completely missing fruits inside or under the canopy.

None of the reviewed papers provide a mathematical or empirical visibility correction factor $(\text{occ})to in fertotalstructura$ from visible fruit counts (vis).

5.3 The Disconnect Between Theoretical Economic Models and Field Observations

On the economic side of the literature, market forecasting models (Mahajan et al., 2026), contract farming game theory (Nematollahi et al., 2025), and lease pricing real-options models (Abdurakhman et al., 2025) formulate sophisticated financial decision frameworks. However, all three economic models rely on abstract mathematical probability distributions (Poisson yield λ, Gaussian yield ξ) or past price lags $(\text{t-1}).Noneoftheseeconomicmodelsincorporatereal-timecomputervisionfielddataorphysicalon-treeobservations.$

5.4 Summary Synthesis and Gap Formulation

Table 1 presents a comprehensive technical comparison across all seven reviewed studies, illustrating their specific capabilities, outputs, estimation levels, and unsolved technical gaps.

Formulated Research Gap Statement Based on this synthesis, the core research gap is formulated as follows:

> *Within the literature represented by these primary sources, there is an absence of an integrated computer vision framework that converts localized 2D/3D visual field observations of pomegranates (visible fruit counts and RGB-D physical diameters) into calibrated tree-level biomass (kg/tree) and total pre-harvest orchard production forecasts while explicitly compensating for foliage occlusion and quantifying error propagation along the pipeline.*

6 Proposed System Architecture and Evaluation Protocol

6.1 Problem Statement and Core Objectives

To bridge the gap between visual sensing and production assessment, this project proposes a Python-based pre-harvest pomegranate yield estimation framework. The core research problem is defined as: *How to mathematically model, correct for foliage occlusion, and aggregate visible fruit detections and RGB-D physical size measurements captured during orchard row traversals into reliable tree-level biomass forecasts and total pre-harvest production estimates with statistical error bounds.*
The primary research objectives are:

Table 1: Technical Comparison of the Seven Reviewed Studies Against Project Scope Requirements.

Study & Reference	Primary Method	Primary Output	Estimation Level	What it Solves	Core Unsolved Gap
Naseer et al. (2024)					
CRnet (2D-CNN + Random Forest)	5-class growth stage label	Cropped organ	98.0% growth stage classification accuracy	No object detection, no counting, no 3D sizing, no yield.	YOLO-Granada (ShuffleNetv2 + CBAM)
Zhao et al. (2024)					
2D bounding box & stage label	2D Frame	Real-time mobile detection (7.74 MB, 29.8 FPS)	No 3D depth/scale, no temporal tracking, no de-duplication, no yield.	Mask R-CNN + LP Tracking + XGBoost	Harvest date remaining days
Giménez-Gallego et al. (2025)					
Single fruit time series	On-tree fruit tracking; harvest date prediction (3.5d error)	Pixel size uncalibrated to cm; top-view misses inner canopy; no tree yield.	DeepLabv3+ + 3D Depth Projection	Physical polar/equatorial diameter (cm)	Exposed fruit
Devanna et al. (2025)					
Calibration-free 3D fruit sizing in field (MAE ~ 1.0 cm)	Discards occluded fruits; no mass (g) conversion; no tree counting or yield.	Tree Ensembles (RF/XGB) vs Naive	1-month-ahead modal price ($\hat{P}_t$)	Mandi / Regional	Benchmarked price forecasting; proved 1-m Naive baseline superiority
Mahajan et al. (2026)					
Excludes exogenous crop yield data; no computer vision or field sensing.	Stackelberg Game Theory (α_{sc})	Optimal fresh volume, wholesale price	Farm / Supply Chain	Proved 17% profit gain for win-win cost-sharing contracts	Uses assumed Gaussian yield (ξ); no computer vision or field sensing.
Nematollahi et al. (2025)					Abdurakhman et al. (2025)
Real Options FTD & DTD Valuation	Fair tree lease price, tenant profit	Tree / Farm	Dynamic lease pricing under low yield threshold risk	Uses assumed Poisson yield (λ); no computer vision or field sensing.	

1. Develop a multi-view video de-duplication and counting pipeline along tree rows using deep learning instance segmentation coupled with temporal multi-target tracking.
2. Formulate and calibrate a 3D volumetric mass allometric model mapping RGB-D projected equatorial $()andpolar()$ diameters to physical fruit mass in grams.
3. Design a canopy visibility correction model (occ)$basedonfoliagedensitymetricstoestimatenon-visibleinner-canopyfruitload(_{total} = C_{occ} \cdot N_{vis})$.
4. Build a spatial yield aggregation pipeline that computes total tree biomass (kg/tree) and orchard production (tons/ha) accompanied by 95% confidence intervals via Monte Carlo error propagation.

6.2 Proposed Python-Based System Architecture

Figure 6 illustrates the proposed end-to-end system architecture comprising six sequential processing modules.

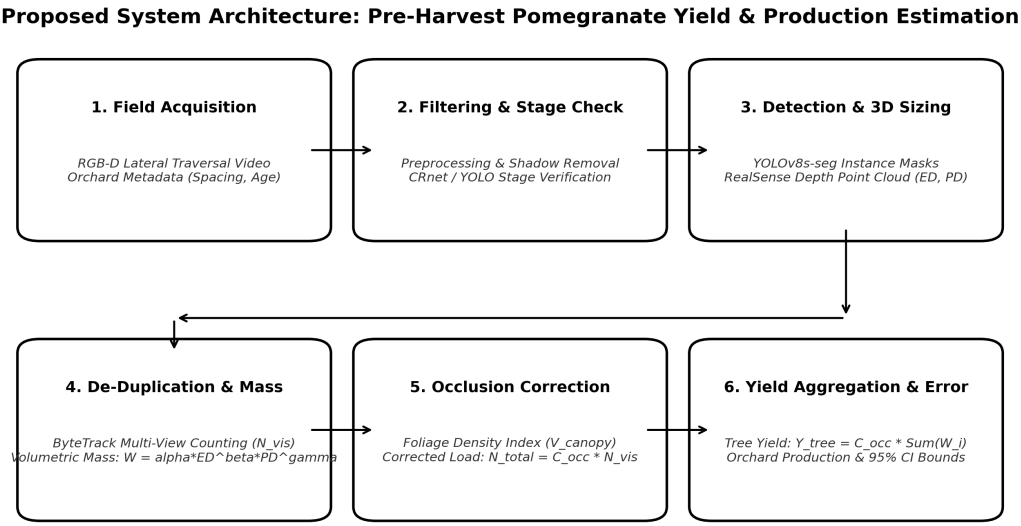


Figure 6: Proposed System Architecture for Python-Based Pre-Harvest Pomegranate Yield Estimation and Production Assessment.

1. **Module 1: Field Video Acquisition & Preprocessing:** Lateral row traversal capturing aligned RGB-D video streams (Intel RealSense D435) at 6 FPS. Contrast-Limited Adaptive Histogram Equalization (CLAHE) is applied to neutralize canopy shadows.
2. **Module 2: Stage Verification & Instance Segmentation:** YOLOv8s-seg generates 2D bounding boxes and instance masks. A lightweight stage classifier (CRnet protocol) filters out unviable buds and flowers, retaining only maturing fruits.
3. **Module 3: 3D Physical Sizing:** Fits 2D circumscribed ellipses to fruit masks and projects vertices onto aligned 3D depth point clouds, applying σ depth std-dev filtering to compute physical equatorial $(, cm)andpolar(,$ cm) diameters (Devanna et al. baseline).

4. **Module 4: Multi-View Tracking & Mass Allometrics:** ByteTrack assigns persistent IDs across video frames to prevent double-counting, yielding non-redundant visible count vis.$Eachvisiblefruit's3Ddimensio$ g) via volumetric allometrics.

5. **Module 5: Canopy Occlusion Correction:** Calculates canopy foliage density index (canopy)$fromgreenleafpixelr$ $1/V_{canopy}$ estimates total fruit load $\hat{N}_{total} = C_{occ} \cdot N_{vis}$.

6. **Module 6: Biomass Aggregation & Error Propagation:** Aggregates individual fruit mass to tree yield ($\hat{Y}_{tree} = C_{occ} \sum \hat{W}_i$) and extrapolates across orchard rows to forecast total production ($\hat{Y}_{total}$, tons/ha) with 95% bootstrap confidence bounds.

6.3 Mathematical Formulation and Volumetric Mass Allometrics

Physical fruit mass ($\hat{W}_i$, grams) is modeled as a function of projected equatorial diameter (, $cm)andpolardiameter($, cm) using a power-law volumetric allometric equation:

$$\hat{W}_i = \alpha \cdot (ED_i)^{\beta} \cdot (PD_i)^{\gamma} \tag{1}$$

where α, β, γ are physical shape and density parameters fitted via non-linear least squares regression on field calibration samples. Assuming an approximate prolate spheroid fruit geometry (= $4\frac{}{3\pi\left(\frac{ED}{2}\right)^2\left(\frac{PD}{2}\right)}$), Equation 1 simplifies to:

$$\hat{W}_i = \rho \cdot \frac{\pi}{6} \cdot (ED_i)^2 \cdot (PD_i) \tag{2}$$

where ρ represents apparent fruit mass density (g/cm^3).
Total estimated crop biomass for a single tree ($\hat{Y}_{tree}$, kg) is expressed as:

$$\hat{Y}_{tree} = \frac{C_{occ}}{1000} \sum_{i=1}^{N_{vis}} \hat{W}_i = \frac{1}{1000 \cdot V_{canopy}} \sum_{i=1}^{N_{vis}} \left[\alpha \cdot (ED_i)^{\beta} \cdot (PD_i)^{\gamma}\right] \tag{3}$$

Total expected orchard production ($\hat{Y}_{total}$, metric tons) across $treesinanorchardblockisaggregatedas : \hat{Y}_{total} = \frac{1}{1000} \sum_{m=1}^{M} \hat{Y}_{tree,m}$(4)

6.4 Data Requirements and Ground-Truth Acquisition

- **Open-Source Pre-training Data:** Mendeley Data pomegranate collection (5,857 images) for pre-training detection and stage classification backbones.

- **Custom Field Video Dataset:** 500 RGB-D image pairs captured across 20 commercial pomegranate trees ('Bhagwa' variety) in Maharashtra, India.

- **Mass Calibration Ground Truth:** Destructive harvest sample of 120 fruits measured for digital caliper diameters (gt, PD$_{gt}$, mm) and digital scale mass (gt, $grams)tofitEquation$ 1.**Tree Yield Ground Truth:**$Totalmanual$ and total harvested fruit mass (actual, $kg/tree)recordedindividuallyfor12designatedground-truthtesttrees.$

6.5 Validation Protocol, Evaluation Metrics, and Error Reporting

System performance will be evaluated across three validation tiers:

1. **Fruit Detection & Sizing:** Precision, Recall, F1-score, mAP@0.5 for detection; MAE, RMSE, and MAPE (%) for 3D diameter sizing against caliper ground truth.

2. **Mass Allometrics:** Coefficient of determination (^{2}), MAE (grams), and RMSE (grams) for individual fruit weight prediction.

3. **Tree Yield & Orchard Production:** Tree yield MAE (kg/tree), MAPE (%), and total production error (%). Statistical error propagation will be reported using non-parametric Monte Carlo bootstrapping (1,000 iterations) to output 95% confidence intervals ($\hat{Y}_{total} \pm \Delta Y$).

6.6 Scope Boundaries, Implementation Roadmap, and Future Work

To maintain technical feasibility within an M.Tech timeline, project implementation is structured into explicit scope tiers:

- **Must-Have Core Deliverables [M.Tech Scope]:** Instance segmentation (YOLOv8s-seg), 3D depth-projected diameter sizing (Devanna protocol), calibrated mass allometric modeling (= f(ED, PD)), $tree-levelyieldaggregation, anddirectbaselinecomparisonagainstuncorrected2Dcounting$.**Good-to-Have Extensio** $viewtemporaltracking(ByteTrack)forrow-levelde-duplication, empiricalcanopyvisibilitycorrectionfacto$ and mobile C++/NCNN export.

- **Future Work (Beyond Scope):** Integration with UAV aerial canopy mapping, and feed-forwarding visual yield predictions into downstream real-options lease pricing models (Abdurakhman et al., 2025) or supply-chain contract games (Nematollahi et al., 2025).

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